

SYSTEM ARCHITECTURE

DRVN 系統架構

由畢業專題的單鏡頭動作分析延伸，整合重訓、跑步、營養三個訓練系統的 App。



A 資料流程：動作分析 POSE ANALYSIS



分工 程式大多在 AI 協助下完成；系統架構、資料流程、驗證方法與檢查標準由我設計。

DATA FLOW

資料流程

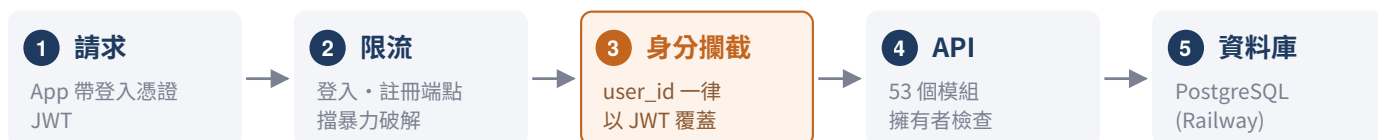
B 訓練課表：生成、紀錄與輪換 TRAINING PLAN



C Apple Watch 跑步 WATCH · RUNNING



D 後端請求 API REQUEST



E 前端打包進 iPhone BUILD



GRADUATION PROJECT · SINGLE CAMERA FITNESS MOTION ANALYSIS

Viewpoint-Specific Capability in Single-Camera Motion Analysis

A three-stage validation of measurability, reliability and discriminative validity: squat and bench press
MediaPipe Pose landmarks · DTW / DBA viewpoint-specific template construction

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Repetition-count accuracy
87%
49 / 48 clips matched the manual count exactly

Drop in the target metric itself
-50%
Squat knee valgus · Frontal 0°: 99.2 ± 49.0 (single-metric score)

Metrics entering the weighted score
23 / 42
42 = measurable 33 = reliable 29 = scored; every exclusion is reported with its reason

Validated capabilities
6.12
viewpoint · fault cells passing discriminative validity: squat 4, bench 2

ABSTRACT

Background Commercial home-fitness systems (e.g. Tempo, Tonali) already output form scores, but most rely on **dedicated hardware**, **depth sensing** or **multiple views** to secure data observability, and few state the conditions under which their measurements remain valid. The user therefore cannot tell whether a low score means the **movement really was poor** or the **camera simply could not see that fault** – the latter creates a false sense of safety. This study deliberately restricts the input to a **single-camera camera**.

Methods Because a single camera offers little observational redundancy, the pipeline is deliberately conservative. **Squat and bench press** were each recorded from three viewpoints; fixed viewpoints were physically set and verified with a **goniometer**, while the self-selected viewpoint angle was recovered post hoc by shoulder-hip projected shortening self-calibration (47°). MediaPipe Pose supplies 33 landmarks, converted into 7 metrics per exercise. Cycles are segmented by a hysteresis state machine with multi-angle consensus voting, after phase reassignment, curves are aligned by **DTW** and averaged by **DBA** into a viewpoint-specific template, penalizing only the pre-defined adverse direction. Validation proceeds through **measurability** → **reliability** → **discriminative validity**.

Results An iPhone prototype was completed. **12 template clips** (squat 6, bench 6) and **46 test clips** (squat 26 incl. 6 off-axis, bench 20) were recorded, with $n=3$ standard and $n=1$ induced-fault trials per cell, each **repetition-count agreement was 87% (40/46)**. Of 42 viewpoint · metric pairs, **33 were measurable** + **29 reliable** + **29 scored** (squat 21, bench 17/21). Of 21 viewpoints · fault cells, **6 were detectable**, **4 unmeasurable**, **2 below threshold** (squat 4/1, bench 2/1/3); thresholds of 15/20/25 give identical verdicts.

Contribution This three-stage framework makes the system **declare the boundary of its own measurement capability** first, turning a single score into a **viewpoint-specific capability map**. **Viewpoint** is the decisive factor for the **capability ceiling**; knee valgus falls from ~50 at frontal 0° to ~2 at self-selected 47°, and is **unmeasurable** at sagittal 90°. **Squat depth** passes at all three. **Bench press** is the mirror image – **elbow flare** is most sensitive over +48°, grip width is measurable only at the self-selected view (+80°). **Limitations** = 1 induced-fault trial per cell, one participant per group.

01 Motivation

Commercial home-fitness systems (e.g. Tempo, Tonali) can indeed score movement quality, but they mostly rely on **dedicated hardware**, **3D depth sensors** or an **extra camera view** to stay stable observation conditions. This study tightens the constraint to one **smartphone camera**. Under that hardware limit, an overlooked question appears: **does a low score mean the movement was genuinely poor**, or **is this viewpoint simply cannot see where you went wrong**? The latter is far more dangerous – the user believes they did it right.

The objective is therefore **not scoring alone**, but two things: ① a deployable single-camera processing pipeline; ② a **measurability-reliability-discriminative validity** three-stage validation that **declines which exercise, at which viewpoint, can reliably judge which fault** – a **viewpoint-specific capability map**.

02 System Pipeline

Following from the motivation, the input is a **single smartphone camera** only – no depth, no multiple views – so the rigour is placed in **processing and validation** cycle segmentation – phase reassignment – DTW alignment – DBA synthesis – three-stage validation, ending in a **viewpoint-specific capability map**. The test exercises are **squat and bench press**, each recorded from three viewpoints.

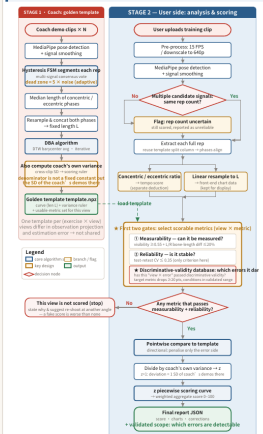


Fig. 1. Two-stage architecture. Stage 1 synthesizes a golden template by DBA and uses the discriminative validity as the scoring view. Stage 2 then compares metrics through measurability and reliability, then applies the pre-established discriminative validity to output a score together with the validated score of one.

03 Measurement Definitions

① **Body-referenced mediolateral axis**
Left-right distances in the image change with body orientation, so we first define:
$$d_{ml} = |P_{lhip} - P_{rhip}|$$
 with the vertical component removed, divided by its own length

making it parallel to the ground and a unit vector. All distances are then computed as the projection onto d_{ml} , giving the participant's own mediolateral width rather than on apparent on-screen distance.
② **Normalisation to a dimensionless ratio** Absolute lengths depend on stature and camera distance, so every distance is divided by a reference length from the same frame (inter-ankle distance for knee valgus, thigh length for depth). The scale factor cancels on division, reducing differences due to body size and shooting distance.

06 Conclusions & Outlook

This study delivers a **deployable single-camera motion-analysis prototype** and establishes an **empirical boundary for measurability quality and fault-detection capability**. **Viewpoint** is not a shooting tip; it is the **experimental factor that sets the system's ceiling**: the drop in the knee-valgus metric itself goes frontal 0° → 50% (detected) → self-selected 47° → 2 pts (below threshold) → sagittal 90° **unmeasurable**. Contributions ① a deployable two-stage system; ② a ground-truth-free surrogate for coordinate boundaries; with the validated scope of use bound to image-estimated relative angle.
Limitations ① $n=1$ induced-fault trial per cell (standard $n=3$), so no effect sizes or confidence intervals are estimated; the results show a consistent trend but still require a larger sample and independent participants. ② One-participant per group. ③ Reliability was measured on the standard group itself. ④ Angles are 2D image estimates, and 50–90° were not densely sampled.

09 Transferability

The validation framework transfers beyond this task. “Measurability-reliability-discriminative validity” is tied neither to fitness nor to MediaPipe – any system that combines monocular vision + measurement + advice faces the same question: on what grounds does it trust what it measured?
Transfer domain The isomorphic problem
Rehabilitation movement assessment Home rehab is unsupervised, “performed incorrectly yet passed” directly affects outcome
Fall-risk screening in elder care Sit-to-stand and gait tests have clinical scales, but single-camera reliability and validity are undefined
Occupational ergonomics Lifting posture assessment (REBA / RULA) needs angle measurement and is still scored by hand
General vision-language models (VLM) as movement coaches They produce confident natural-language judgements but will not say “I cannot see this from this angle” – this study measured exactly that failure (valgus information disappears at 90°), so the framework is an **evaluation protocol** they need too

REFERENCES

[1] Dima, A. M. Reliability and concurrent validity between 2D and 3D evaluations of knee valgus during drop jumps. *Open Access J Sports Med* 2018;10:75–79. [2] Strath, M. Powers, C. M. A biomechanical view of the squat exercise. *Int J Sports Phys Ther* 2024;19:490–501. [3] Bazzucchi, V. et al. Real-time on-device real-time body pose tracking. *arXiv:2601.10204*, 2026. [4] Saito, H. Chiba, S. Dynamic programming algorithm optimization for system wide recognition. *Euro Trans ASIP* 2019;20:43–49. [5] Pothier, J. et al. A global 3D pose estimation for dynamic time warping. *Pattern Recognit* 2021;142:1679–693. [6] Savitsky, G. M. Smoothing and differentiation of data by smoothed fast square penalties. *Ann Chem* 1964;36:1627–439. [7] Hopkins, W. Measures of reliability in sports medicine. *Sports Med* 2002;33(1):1–15.

Reint · Capacitor (IOS) · FastAPI · Railway · MediaPipe Pose · SciPy · tableau · Smoothing: Savitzky-Golay filter
All four decision thresholds (0.35 / 20% / 0.35 / 20 pts) were derived in this study's pilot and declared in advance, not taken from the literature | The measurement validity of the original KASR is cited from [1]; this study uses a MediaPipe landmark-adapted formulation

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github.com/148981/
drvn-portfolio